

Math 521, Section 2
Practice Final Exam
Spring 2026

First name: _____ **Last name:** _____

Instructions:

- This exam contains 8 problems, and there are a total of 80 points available.
- Show all your work in the space provided. You may also use the backs of pages.
- No outside resources are allowed, including notes, calculators, textbooks, etc.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
Total:	80	

1. (10 points) Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be functions. Prove that if $(g \circ f)(a) = a$ for all $a \in A$ and $(f \circ g)(b) = b$ for all $b \in B$, then f is bijective.

2. Suppose that $\{a_n\}_{n \geq 1}$ is a sequence of real numbers that converges, and let $m \in \mathbb{R}$.

Prove or disprove the following statements:

- (a) (5 points) If $a_n \geq m$ for all $n \geq 1$, then $\lim_{n \rightarrow \infty} a_n \geq m$.
- (b) (5 points) If $a_n > m$ for all $n \geq 1$, then $\lim_{n \rightarrow \infty} a_n > m$.

3. (a) (6 points) Using induction, prove that for all $n \in \mathbb{N}$ we have

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}.$$

- (b) (4 points) Determine if the series

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

converges and prove your answer.

4. (10 points) Let $\sum_{n=1}^{\infty} a_n$ be a series of real numbers. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges.

5. Consider the metric space $(\mathbb{R}^2, d_2)$ and the subspace

$$Y = \{(x, y) \in \mathbb{R}^2 : y > 0\}$$

with the induced metric d_Y .

- (a) (5 points) Prove that for any $A \subseteq Y$, if A is open in (Y, d_Y) then A is open in $(\mathbb{R}^2, d_2)$.
- (b) (5 points) Provide an example of a set $B \subseteq Y$ where B is closed in (Y, d_Y) and not closed in $(\mathbb{R}^2, d_2)$, and prove your answer.

6. Consider the metric space $(\mathbb{R}, |\cdot|)$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. **Prove or disprove** the following statements:
- (a) (5 points) If f is continuous, then $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq \mathbb{R}$.
 - (b) (5 points) If f is continuous, then $f(\overline{A}) \supseteq \overline{f(A)}$ for any $A \subseteq \mathbb{R}$.

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7. (10 points) Let X and Y be metric spaces and $f : X \rightarrow Y$ be a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \rightarrow X$ is also continuous.

8. Consider the power series $\sum_{n=1}^{\infty} \sqrt{n} x^n$.

- (a) (4 points) Find its radius of convergence R .
- (b) (6 points) For each $x \in \mathbb{R}$, determine if the series converges and prove your answer.

Extra paper
